

```
!pip install sentence-transformers
!pip install torch torchvision
!pip install faiss-cpu
```

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Requirement already satisfied: sentence-transformers in
/usr/local/lib/python3.12/dist-packages (5.5.1)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(5.10.1)
Requirement already satisfied: huggingface-hub>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(1.17.0)
Requirement already satisfied: torch>=1.11.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(2.11.0+cu128)
Requirement already satisfied: numpy>=1.20.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(2.0.2)
Requirement already satisfied: scikit-learn>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(1.6.1)
Requirement already satisfied: scipy>=1.0.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(1.16.3)
Requirement already satisfied: typing_extensions>=4.5.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(4.15.0)
Requirement already satisfied: tqdm>=4.0.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(4.67.3)
Requirement already satisfied: click>=8.4.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0-
>sentence-transformers) (8.4.1)
Requirement already satisfied: filelock>=3.10.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0-
>sentence-transformers) (3.29.1)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0-
>sentence-transformers) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0-
>sentence-transformers) (1.5.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0-
>sentence-transformers) (0.28.1)
Requirement already satisfied: packaging>=20.9 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0-
>sentence-transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0-
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>sentence-transformers) (6.0.3)
Requirement already satisfied: typer<0.26.0,>=0.20.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0-
>sentence-transformers) (0.25.1)
Requirement already satisfied: joblib>=1.2.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.22.0-
>sentence-transformers) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.22.0-
>sentence-transformers) (3.6.0)
Requirement already satisfied: setuptools<82 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (3.1.6)
Requirement already satisfied: cuda-toolkit==12.8.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenv,
nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (12.8.1)
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (12.9.7)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (9.19.0.56)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (3.4.5)
Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (3.6.0)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenv,
nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
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```
>sentence-transformers) (12.8.4.1)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.*
in /usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (12.8.90)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (11.3.3.83)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (1.13.1.3)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (12.8.90)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (11.7.3.90)
Requirement already satisfied: nvidia-cuspars
e-cu12==12.5.8.93.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (12.5.8.93)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (12.8.93)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (12.8.93)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (12.8.90)
Requirement already satisfied: nvidia-cuda-nvtx-cu12==12.8.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars
e,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (12.8.90)
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tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.11.0-
>sentence-transformers) (12.8.90)
Requirement already satisfied: regex>=2025.10.22 in
/usr/local/lib/python3.12/dist-packages (from
transformers<6.0.0,>=4.41.0->sentence-transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from
transformers<6.0.0,>=4.41.0->sentence-transformers) (0.22.2)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from
transformers<6.0.0,>=4.41.0->sentence-transformers) (0.7.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-
bindings<13,>=12.9.4->torch>=1.11.0->sentence-transformers) (1.5.5)
Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub>=0.23.0->sentence-transformers) (4.13.0)
Requirement already satisfied: certifi in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub>=0.23.0->sentence-transformers) (2026.5.20)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub>=0.23.0->sentence-transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub>=0.23.0->sentence-
transformers) (3.18)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface-hub>=0.23.0->sentence-transformers)
(0.16.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3-
>torch>=1.11.0->sentence-transformers) (1.3.0)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer<0.26.0,>=0.20.0-
>huggingface-hub>=0.23.0->sentence-transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in
/usr/local/lib/python3.12/dist-packages (from typer<0.26.0,>=0.20.0-
>huggingface-hub>=0.23.0->sentence-transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer<0.26.0,>=0.20.0-
>huggingface-hub>=0.23.0->sentence-transformers) (0.0.4)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0-
>sentence-transformers) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=13.8.0-
>typer<0.26.0,>=0.20.0->huggingface-hub>=0.23.0->sentence-
transformers) (4.2.0)
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=13.8.0-
>typer<0.26.0,>=0.20.0->huggingface-hub>=0.23.0->sentence-
transformers) (2.20.0)

Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=13.8.0->typer<0.26.0,>=0.20.0->huggingface-hub>=0.23.0-
>sentence-transformers) (0.1.2)

Requirement already satisfied: torch in
/usr/local/lib/python3.12/dist-packages (2.11.0+cu128)

Requirement already satisfied: torchvision in
/usr/local/lib/python3.12/dist-packages (0.26.0+cu128)

Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from torch) (3.29.1)

Requirement already satisfied: typing-extensions>=4.10.0 in
/usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)

Requirement already satisfied: setuptools<82 in
/usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)

Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)

Requirement already satisfied: fsspec>=0.8.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)

Requirement already satisfied: cuda-toolkit==12.8.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch) (12.8.1)

Requirement already satisfied: cuda-bindings<13,>=12.9.4 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.9.7)

Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in
/usr/local/lib/python3.12/dist-packages (from torch) (9.19.0.56)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in
/usr/local/lib/python3.12/dist-packages (from torch) (2.28.9)

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.4.5)

Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.6.0)

Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in
/usr/local/lib/python3.12/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)
(12.8.4.1)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.*
in /usr/local/lib/python3.12/dist-packages (from cuda-

```
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvj  
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)  
(12.8.90)
```

```
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in  
/usr/local/lib/python3.12/dist-packages (from cuda-  
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvj  
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)  
(11.3.3.83)
```

```
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in  
/usr/local/lib/python3.12/dist-packages (from cuda-  
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvj  
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)  
(1.13.1.3)
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Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in  
/usr/local/lib/python3.12/dist-packages (from cuda-  
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvj  
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)  
(12.8.90)
```

```
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in  
/usr/local/lib/python3.12/dist-packages (from cuda-  
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvj  
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)  
(10.3.9.90)
```

```
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in  
/usr/local/lib/python3.12/dist-packages (from cuda-  
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvj  
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)  
(11.7.3.90)
```

```
Requirement already satisfied: nvidia-cusparsenvj  
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)  
(12.5.8.93)
```

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Requirement already satisfied: nvidia-nvj  
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)  
(12.8.93)
```

```
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in  
/usr/local/lib/python3.12/dist-packages (from cuda-  
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvj  
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)  
(12.8.93)
```

```
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in  
/usr/local/lib/python3.12/dist-packages (from cuda-  
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvj  
tlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)  
(12.8.90)
```

```
Requirement already satisfied: numpy in
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/usr/local/lib/python3.12/dist-packages (from torchvision) (2.0.2)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in
/usr/local/lib/python3.12/dist-packages (from torchvision) (11.3.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-
bindings<13,>=12.9.4->torch) (1.5.5)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch)
(1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Collecting faiss-cpu
  Downloading faiss_cpu-1.14.2-cp310-abi3-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (7.8 kB)
Requirement already satisfied: numpy>=1.25 in
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (2.0.2)
Requirement already satisfied: packaging in
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (26.2)
Downloading faiss_cpu-1.14.2-cp310-abi3-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (18.2 MB)
18.2/18.2 MB 99.0 MB/s eta
0:00:00

```

```
import pandas as pd
```

```
df = pd.read_csv("products.csv")
```

```
print(df.head())
```

```
print(df.columns)
```

	product_id	brand	\
0	B08YRWN3WB	JANSPORT	
1	B08YRXFZZM	JANSPORT	
2	B09Q2PQ7ZB	BAODINI	
3	B001BEAWXY	Calvin Klein	
4	B09FM5PMN3	JANSPORT	

	category	\	title	price	
0	Big Student Large laptop backpack	Black	EK0A5B...	189.00	New season
1			Superbreak Day Pack	119.00	New season
2	Mini Travel Umbrella With Case	Small Compact U...		17.79	New season
3		Men's Cotton Classics 3-Pack	Boxer Brief	119.00	New season
4			Unisex Kids Bag Backpack	179.00	New season

```

    rating                                     image_url \
0      4.7  https://m.media-amazon.com/images/I/51y2EF00m0...
1      4.6  https://m.media-amazon.com/images/I/51yvvQUs3S...
2      4.2  https://m.media-amazon.com/images/I/71WbrZPbnG...
3      4.2  https://m.media-amazon.com/images/I/716vzeqa7Z...
4      4.4  https://m.media-amazon.com/images/I/51XASBuqjr...

                                     product_url
0  https://www.amazon.ae/dp/B08YRWN3WB
1  https://www.amazon.ae/dp/B08YRXFZZM
2  https://www.amazon.ae/dp/B09Q2PQ7ZB
3  https://www.amazon.ae/dp/B001BEAWXY
4  https://www.amazon.ae/dp/B09FM5PMN3
Index(['product_id', 'brand', 'title', 'price', 'category', 'rating',
      'image_url', 'product_url'],
      dtype='object')

print(df.columns)

Index(['product_id', 'brand', 'title', 'price', 'category', 'rating',
      'image_url', 'product_url'],
      dtype='object')

print(df.shape)

(13156, 8)

df.isnull().sum()

product_id      0
brand           1
title           1
price          193
category        0
rating         883
image_url       0
product_url     0
dtype: int64

df["brand"] = df["brand"].fillna("Unknown")
df["title"] = df["title"].fillna("Unknown")

df["price"] = df["price"].fillna(df["price"].median())
df["rating"] = df["rating"].fillna(df["rating"].median())

df.duplicated().sum()

np.int64(667)

df = df.drop_duplicates()

```



```

df.duplicated().sum()
np.int64(0)
df["title"] = df["title"].str.strip()
df["title"] = df["title"].str.lower()
print(df.shape)
(12489, 8)
import re

def clean_text(text):
    text = re.sub(r'^a-zA-Z0-9 ', '', str(text))
    return text

df["title"] = df["title"].apply(clean_text)
print(df["brand"].nunique())
2409
print(df["brand"].value_counts().head())
brand
Casio      484
adidas     478
Skechers   436
Nike       288
Tommy Hilfiger 251
Name: count, dtype: int64

df.isnull().sum()
product_id    0
brand         0
title         0
price         0
category      0
rating        0
image_url     0
product_url   0
dtype: int64

df["combined_text"] = (
    df["brand"] + " " +
    df["title"] + " " +
    df["category"]
)

df.to_csv("cleaned_amazon_fashion.csv", index=False)

```

```

from sentence_transformers import SentenceTransformer

model = SentenceTransformer('all-MiniLM-L6-v2')

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
  warnings.warn(

{"model_id": "dcc944c7d8f74a399027466cdbba2ab3", "version_major": 2, "version_minor": 0}

Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.

{"model_id": "206bab3bae7647609598eeab94e57cda", "version_major": 2, "version_minor": 0}

{"model_id": "1e4b00a44ff44811b39289063cdab501", "version_major": 2, "version_minor": 0}

{"model_id": "ba56cc57021f4693a770be530281a928", "version_major": 2, "version_minor": 0}

{"model_id": "f186e9bf27f34ef4bd0375c06456a6b4", "version_major": 2, "version_minor": 0}

{"model_id": "2807982e951f4c5b94a9577c067f229a", "version_major": 2, "version_minor": 0}

{"model_id": "965d8789e6574a30b70a82e4e6d9b910", "version_major": 2, "version_minor": 0}

{"model_id": "49f4ccc97b8d49ee888d4c430d2d7567", "version_major": 2, "version_minor": 0}

{"model_id": "5f94e1ad82c94708a824fbcaa26bb91f", "version_major": 2, "version_minor": 0}

{"model_id": "bbc7cd83bc4148af8dc3b971dece7c7c", "version_major": 2, "version_minor": 0}

```

```

{"model_id": "d66cfd473ee443be93fca7f4a2ffclea", "version_major": 2, "version_minor": 0}

{"model_id": "67d549437401489b9bd559e7c6198af5", "version_major": 2, "version_minor": 0}

text_embeddings = model.encode(
    df["combined_text"].tolist(),
    show_progress_bar=True
)

{"model_id": "ea21d71b4e7044baafdeaa7619a79483", "version_major": 2, "version_minor": 0}

print(text_embeddings.shape)

(12489, 384)

from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df["brand_encoded"] = le.fit_transform(df["brand"])

from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
df["price_scaled"] = scaler.fit_transform(
    df[["price"]]
)

import numpy as np

brand_feature = df["brand_encoded"].values.reshape(-1, 1)
price_feature = df["price_scaled"].values.reshape(-1, 1)

final_features = np.hstack([
    text_embeddings,
    brand_feature,
    price_feature
])

print(final_features.shape)

(12489, 386)

from sklearn.preprocessing import normalize
final_features = normalize(final_features)

```

```

#similarity check with text,brand,price
from sklearn.metrics.pairwise import cosine_similarity
query = final_features[10].reshape(1, -1)

similarities = cosine_similarity(
    query,
    final_features
)

print(similarities.shape)

(1, 12489)

import numpy as np

similar_indices = np.argsort(
    similarities[0]
)[::-1]

print(similar_indices[:5])

[ 10  18 5087  27 132]

def recommend(product_index, top_n=5):
    query = final_features[
        product_index
    ].reshape(1,-1)

    similarities = cosine_similarity(
        query,
        final_features
    )

    similar_products = np.argsort(
        similarities[0]
    )[::-1]

    recommendations = similar_products[
        1:top_n+1
    ]

    return recommendations

recommendations = recommend(10)

for idx in recommendations:
    print(
        df.iloc[idx]["title"]
    )

```

```

unisex elmntl hbr backpack
mens elmntl lbr backpack
unisex elmntl prm backpack
unisex nk elmntl prm bkpk backpack
mens air grx backpack

!pip install faiss-cpu

Collecting faiss-cpu
  Downloading faiss_cpu-1.14.2-cp310-abi3-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (7.8 kB)
Requirement already satisfied: numpy>=1.25 in
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (2.0.2)
Requirement already satisfied: packaging in
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (26.2)
Downloading faiss_cpu-1.14.2-cp310-abi3-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (18.2 MB)
----- 18.2/18.2 MB 87.0 MB/s eta
0:00:00

#FAISS vector search library sys from meta
import faiss
dimension = final_features.shape[1]

index = faiss.IndexFlatIP(
    dimension
)

index.add(
    final_features.astype(
        "float32"
    )
)

print(index.ntotal)
12489

query = final_features[
    10
].reshape(1, -1)

scores, indices = index.search(
    query.astype("float32"),
    6
)
print(scores)
print(indices)

[[1.      1.      1.      1.      0.99999994 0.99999994]]
[[5087   27   18   10  133  132]]

```

```

for idx in indices[0][1:]:
    print(
        df.iloc[idx]["title"]
    )
print("-----")
print(df.iloc[10]["title"])

unisex elmntl prm backpack
unisex elmntl hbr backpack
mens elmntl lbr backpack
mens air grx backpack
unisex nk elmntl prm bkpk backpack
-----
mens elmntl lbr backpack

#multimodal ie moving text recommendation to image recommendation
from torchvision import models
import torch

resnet = models.resnet50(
    pretrained=True
)

/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:208: UserWarning: The parameter 'pretrained' is deprecated
since 0.13 and may be removed in the future, please use 'weights'
instead.
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:2
23: UserWarning: Arguments other than a weight enum or `None` for
'weights' are deprecated since 0.13 and may be removed in the future.
The current behavior is equivalent to passing
`weights=ResNet50_Weights.IMAGENET1K_V1`. You can also use
`weights=ResNet50_Weights.DEFAULT` to get the most up-to-date weights.
  warnings.warn(msg)

Downloading: "https://download.pytorch.org/models/resnet50-
0676ba61.pth" to /root/.cache/torch/hub/checkpoints/resnet50-
0676ba61.pth

100%|██████████| 97.8M/97.8M [00:00<00:00, 232MB/s]

resnet.fc = torch.nn.Identity()
resnet.eval()

ResNet(
  (conv1): Conv2d(3, 64, kernel_size=(7, 7), stride=(2, 2),
padding=(3, 3), bias=False)
  (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)

```

```

    (relu): ReLU(inplace=True)
    (maxpool): MaxPool2d(kernel_size=3, stride=2, padding=1, dilation=1,
ceil_mode=False)
    (layer1): Sequential(
      (0): Bottleneck(
        (conv1): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (downsample): Sequential(
          (0): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
          (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        )
      )
      (1): Bottleneck(
        (conv1): Conv2d(256, 64, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
      )
      (2): Bottleneck(
        (conv1): Conv2d(256, 64, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)

```

```

        (conv3): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
    )
)
(layer2): Sequential(
  (0): Bottleneck(
    (conv1): Conv2d(256, 128, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(2, 2),
padding=(1, 1), bias=False)
    (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (relu): ReLU(inplace=True)
    (downsample): Sequential(
      (0): Conv2d(256, 512, kernel_size=(1, 1), stride=(2, 2),
bias=False)
      (1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    )
  )
  (1): Bottleneck(
    (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
    (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (relu): ReLU(inplace=True)
  )
  (2): Bottleneck(
    (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)

```



```

        (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
    )
    (3): Bottleneck(
        (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
    )
)
(layer3): Sequential(
  (0): Bottleneck(
    (conv1): Conv2d(512, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(2, 2),
padding=(1, 1), bias=False)
    (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (relu): ReLU(inplace=True)
    (downsample): Sequential(
      (0): Conv2d(512, 1024, kernel_size=(1, 1), stride=(2, 2),
bias=False)
      (1): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    )
  )
)
  (1): Bottleneck(

```

```

        (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
    )
    (2): Bottleneck(
        (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
    )
    (3): Bottleneck(
        (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
    )
    (4): Bottleneck(
        (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)

```

```

        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
    )
    (5): Bottleneck(
        (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
    )
)
(layer4): Sequential(
  (0): Bottleneck(
    (conv1): Conv2d(1024, 512, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(2, 2),
padding=(1, 1), bias=False)
    (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv3): Conv2d(512, 2048, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn3): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (relu): ReLU(inplace=True)
    (downsample): Sequential(
      (0): Conv2d(1024, 2048, kernel_size=(1, 1), stride=(2, 2),
bias=False)
      (1): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    )
  )
)
  (1): Bottleneck(

```

```

        (conv1): Conv2d(2048, 512, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(512, 2048, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
    )
    (2): Bottleneck(
        (conv1): Conv2d(2048, 512, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(512, 2048, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
    )
)
(avgpool): AdaptiveAvgPool2d(output_size=(1, 1))
(fc): Identity()
)

import requests
from PIL import Image
from io import BytesIO

url = df.loc[0, "image_url"]

response = requests.get(url)
image = Image.open(
    BytesIO(response.content)
)

image

```



```
from torchvision import transforms
transform = transforms.Compose([
    transforms.Resize((224,224)),
    transforms.ToTensor()
])

image_tensor = transform(
    image
)

print(image_tensor.shape)
torch.Size([3, 224, 224])

image_tensor = image_tensor.unsqueeze(0)

#generate embedding
with torch.no_grad():
    embedding = resnet(
        image_tensor
    )
```

```

print(
    embedding.shape
)
torch.Size([1, 2048])
embedding = (
    embedding
    .squeeze()
    .numpy()
)
print(
    embedding.shape
)
(2048,)
def image_embedding_from_url(url):
    try:
        response = requests.get(
            url,
            timeout=10
        )
        image = Image.open(
            BytesIO(response.content)
        ).convert("RGB")
        image_tensor = transform(
            image
        ).unsqueeze(0)
        with torch.no_grad():
            embedding = resnet(
                image_tensor
            )
        return (
            embedding
            .squeeze()
            .numpy()
        )
    except:
        return np.zeros(2048)

```

```

emb = image_embedding_from_url(
    df.loc[0, "image_url"]
)

print(emb.shape)

(2048,)

image_embeddings = []

for url in df["image_url"][:100]:
    emb = image_embedding_from_url(
        url
    )
    image_embeddings.append(
        emb
    )

image_embeddings = np.array(
    image_embeddings
)

print(
    image_embeddings.shape
)

(100, 2048)

text_emb_100 = text_embeddings[:100]
brand_emb_100 = brand_feature[:100]
price_emb_100 = price_feature[:100]

print(text_emb_100.shape)
print(brand_emb_100.shape)
print(price_emb_100.shape)
print(image_embeddings.shape)

(100, 384)
(100, 1)
(100, 1)
(100, 2048)

import numpy as np

final_features = np.hstack([
    text_emb_100,
    brand_emb_100,
    price_emb_100,
    image_embeddings
])

```

```

print(final_features.shape)
(100, 2434)
from sklearn.preprocessing import normalize
final_features = normalize(
    final_features,
    norm='l2'
)
from sklearn.metrics.pairwise import cosine_similarity
similarity_matrix = cosine_similarity(
    final_features
)
def recommend(product_idx, top_n=5):
    similarities = similarity_matrix[
        product_idx
    ]
    similar_products = np.argsort(
        similarities
    )[::-1]
    return similar_products[1:top_n+1]
indices = recommend(0)
print("Selected Product:")
print(df.iloc[0]["title"])
print("\nRecommendations:")
for idx in indices:
    print(df.iloc[idx]["title"])
Selected Product:
big student large laptop backpack black ek0a5bahn551
Recommendations:
unisex kids bag backpack
unisex big day pack backpacks
unisexadult main campus backpack
superbreak backpack
large accessory pouch secure storage space for pens power cords
pencil case ideal for everyday need 13l navy
from IPython.display import display, HTML

```



```

indices = recommend(1)

html = "<h3>Selected Product</h3>"
html += f"""
<div style='display:inline-block; margin:10px; text-align:center;'>
    <br>
    <b>{df.iloc[1]['title']}</b>
</div>
"""

html += "<h3>Recommendations</h3>"

for idx in indices:
    html += f"""
    <div style='display:inline-block; margin:10px; text-align:center;
vertical-align:top;'>
        <br>
        <span>{df.iloc[idx]['title']}</span>
    </div>
    """

display(HTML(html))
<IPython.core.display.HTML object>

```